



Deepfake news: AI-enabled disinformation as a multi-level public policy challenge

Christopher Whyte

To cite this article: Christopher Whyte (2020) Deepfake news: AI-enabled disinformation as a multi-level public policy challenge, *Journal of Cyber Policy*, 5:2, 199-217, DOI: [10.1080/23738871.2020.1797135](https://doi.org/10.1080/23738871.2020.1797135)

To link to this article: <https://doi.org/10.1080/23738871.2020.1797135>



Published online: 31 Jul 2020.



Submit your article to this journal [↗](#)



Article views: 5643



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 44 View citing articles [↗](#)



Deepfake news: AI-enabled disinformation as a multi-level public policy challenge

Christopher Whyte

L. Douglas Wilder School of Government and Public Affairs, Virginia Commonwealth University

ABSTRACT

The advent of ‘DeepFake’ content that is increasingly difficult for humans and machines to distinguish as artificial portends a number of challenges to democratic societies. In order to effectively respond, policymakers must gain understanding of how DeepFake content might manifest. This paper aims to offer necessary context by exploring AI-enabled multimedia disinformation across different levels: (1) as a mass-produced, regular feature of the information environment in democracies and (2) as a highly tailored instrument used in tandem with cyber operations. I explore the impact of DeepFakes on the ability of populations to determine the origination, credibility, quality and freedom of information. Such macro impacts amplify the potential value of DeepFake content employed alongside targeted cyber activities, a combination that even alone offers belligerent actors new opportunities for enhancing attempts at disinformation and coercion. Nevertheless, I ultimately argue that DeepFakes should be thought of more as an evolution than a revolution in disinformation techniques, the real threat of which emerges from the manner in which new abilities to produce even reasonable fidelity fabrications rapidly and at scale combine the multiform shape of the modern digital information environment to make organized influence efforts much more dynamic than has previously been the case.

ARTICLE HISTORY

Received 2 August 2019

Revised 19 May 2020

Accepted 27 May 2020

KEYWORDS

Disinformation; subversion;
artificial intelligence;
deepfake; cyber operations

1. Introduction

Subversion and fake news are not new features of world politics (Sunstein and Vermeule 2009); in centuries past, sophisticated political actors have regularly employed disinformation for strategic gain. Louis XIV and Ivan the Terrible both adopted foreign policies of subversion in advance of military campaigns in Central and Eastern Europe that aimed to create conditions among local populations favourable to eventual occupation. More recently, the governments of both blocs during the Cold War extensively turned to propaganda and the manipulation of social process in order to manipulate geostrategic conditions. In the Soviet case, their actions in doing so continued at least 150 years of subversion as state practice under the preceding czarist empire (Abrams 2016; Hotchkiss 2017). And such subversion is not only the common and enduring practice of well-resourced state actors. ‘Shallow’ fake news and propaganda that is relatively easy to

distinguish from other content in national information environments has allowed both pillars of society and opponents thereof to shape popular discourse (Spencer and Spencer 2007; Kaplan 2008). Likewise, terrorist actors have regularly used imagery of clearly false providence to cause confusion and capture the eyes of internet users. Recent distribution of fake victim images following suicide bombing attacks in India in early 2019 are just one such example thereof (Kedar 2020).

The challenges of combating disinformation today have consumed the attention of national security and technology experts in stark fashion. However, it is the challenges of tomorrow that are quite arguably the greatest source of fear and fatalism about the proper function of democracies in an age of advanced information technologies. A common refrain of information warfare specialists in just the past couple of years has been (generically phrased): how much worse might Western experiences with broad-scoped fake news and foreign-based attempts at subversion get once the perpetrators of influence operations begin to take fuller advantage of advances in artificial intelligence? High fidelity fabricated multimedia could be used to accuse political candidates of scandalous behaviour or could be cited as a defence for legitimately corrupt or criminal persons. Limited changes to broad bodies of recorded media on elements of a notable figure's past public career could poison the historical record in the eyes of the public, feeding sensational narrative even where no firm evidence can be found. And fabricated video footage could be used to pre-empt or subvert the public-facing position of entities legitimately dealing with secret information, such as law enforcement or intelligence organisations. Simply put, the advent of fabrications that are not only increasingly sophisticated, but also increasingly cheap, diverse in format and simple to develop promises a democratisation of disinformation beyond what has otherwise been a relatively common feature of human history.

The remainder of this article aims to offer necessary context and public policy perspective by exploring AI-enabled disinformation as (1) an everyday foil to the normal information consumption of the average democratic citizen, (2) a highly tailored instrument that can be used to enhance the effect of other criminal or conflict actions, such as cyber operations, and (3) a mass-produced, regular feature of the information environment in democracies. Focusing on the case of the 'DeepFake', I argue that macro impacts on the ability of populations to assess the credibility, origination and reality of information amplifies the potential value of the technology when employed alongside targeted cyber activities. In doing so, I offer a nuanced read of the threat of DeepFakes – that while they should be thought of more as an evolution than a revolution in disinformation techniques, the intersection of abilities to produce even *reasonable* fidelity fabrications (Johnson 2019) rapidly and at scale with the multiform shape of the modern digital information environment stand to make organised influence efforts more dynamic and worrisome than ever before.

2. Fake news and DeepFakes

'Fake news'¹ is information developed and disseminated to mislead the targeted audience for some sociopolitical or economic gain. In the broadest possible sense, one might think that the term means little more than content that contains any degree of *intended* factual inaccuracy, ranging from modified detail to full fabrication. There is some degree of value

to this broad definition insofar as it emphasises directed manipulation of information. But as a number of scholars have pointed out, the term is often employed to imply so many distinct phenomena that it is of limited utility to rigorous empirical research projects that fail to lay out strict definitional parameters (Wardle 2017). Wardle and Derakhshan (2017) note (or reference others' observations) that the term has variously been used by pundits and scholars alike to discuss an immense array of activities in just the past few years, including satire, parody, propaganda, manipulative advertising, false balance (meaning news unworthy of popular focus) and *disinformatya* (meaning information used to sow doubt) (Tandoc, Lim, and Ling 2017; Zuckerman 2017; Boyd 2017; Sullivan 2017; Jack 2017), as well as activities as distinct as misinformation, disinformation and mal-information. Going forward, I simply use the term 'disinformation' to describe the potential use of a single new technology (i.e. DeepFakes) for broad-scoped information manipulation.²

While fake news is not the stuff of just the past several years, Western experiences with widespread disinformation in just the past few years have intensified in unprecedented fashion (Woolley and Howard 2018).³ In the 2016 presidential election in the United States, Russian-backed operatives published more than 82 million distinct messages on sites like Instagram, Facebook and Twitter sharing an estimated 80,000 pieces of fake or modified content (DiResta et al. 2019). In the United States alone, it is thought that about 126 million people were exposed to the direct or subsequent republished versions of that content (Way and Casey 2019; DiResta et al. 2019; Shane and Mazzetti 2018). Elsewhere, Russian efforts to skew the social and political processes of Western states have seen widespread use of defamatory information produced by 'troll farms' and targeted to one end or another.⁴ In the Czech Republic, Italy, Germany and the Netherlands, cyber-enabled disinformation aimed to aid the election of Eurosceptical political parties; in Bulgaria, disinformation sought to aid the campaign of a pro-Russian candidate; and in both the United Kingdom and Spain, fake news was employed to fan the flames of secessionist messaging (Way and Casey 2019, 3–4).

But fake news today is not merely the product of state-sponsored campaigns of societal interference. Populist leaders riding waves of nationalist backlash against progressive political establishments have turned to lean heavily on affective appeals to national audiences that often blur or completely ignore the factual nature of issues under discussion. And national media ecosystems are themselves increasingly polluted with misleading or meaningless content, the existence of which oddly seems both correlated to the increasing incidence of directed disinformation campaigns and yet topically unaffiliated.⁵ At the same time, the convention of the media watchdog is suffering. Even in bastions of democratic process in the West, the conversation about what is 'real' in the context of contemporary political discourse has led to stark variation across the spectrum of news media operators about the validity of 'post-factual' rhetoric.⁶

In exactly this fashion, the current challenges faced by global society with fake news are in some senses no different than those we will face in the future. Only changing tools complicate the task of public policymakers. Today, the pervasiveness of fake news as a foil to traditional methods of assuring information integrity in the media publication and democratic political processes quite arguably comes down to the rise of social networking services based on the internet.⁷ Tomorrow, the prospective pervasiveness thereof is likely to be more based around the use of artificial intelligence applications for developing fabricated content.

2.1. Enter the DeepFake

DeepFakes (a portmanteau of the terms ‘deep learning’ and ‘fake’), which are set to add a layer of complexity to contemporary efforts to safeguard democratic process, are multimedia made to mimic real-world events at such a high level of fidelity so as to be indistinguishable therefrom. DeepFakes make use of machine learning systems called deep neural networks. A neural network is a classification of system that broadly attempts to mimic the way human brains work in order to allow computers to ‘learn’ new behaviours (Beale, Demuth, and Hagan 1996). Traditionally, computers are distinct from the purpose-built machines that preceded the digital age – such as the abacus or water wheel – in that they are reprogrammable. Not only can computers be built to mimic some element of human function (i.e. human labour or cognition); they can also be re-programmed to accomplish other tasks and are thus general-purpose. The issue with computers in this traditional design conception and implementation is that they must rely on human inputs in order to function well. The result is a degree of rigidity not found in humans, which are able to consider related or tangential examples in the world around them, infer and then act in an informed fashion even where there they have no pre-existing approach in mind.⁸ Neural networks aim to mimic this behaviour.

DeepFakes are digital forgery, something that is hardly new in the world. For many years, the ability to manually alter digital documentation or imagery has enabled disinformation and fraud along various lines. Perhaps most straightforwardly, use of software like Photoshop and Adobe Premier might be used to create dopplegangers of real people as a means of attempting to convince an audience that those people have stated, advocated or acted in ways they have not. Such work must be done manually at the cost of some man-hours – scaling depending on the scope and quality of the required task – but are nevertheless found commonly enough online as the result of a targeted effort to entertain, attack or otherwise mimic someone.

Such digital forgery is often far from perfect. However, altered content along such lines is often employed successfully for purposes of disinformation beyond the simple attempt to sell fake news. Altering documentation, particularly prior to initial publication, can challenge the ability of mainstream observers to effectively document history. If a standalone photo of a critical event published by an individual – say, someone who was in the right place at the right time with a phone camera – contains evidence of editing but not perceivable signs of forgery, what is the actual evidentiary value of the image (Gregory n.d.)? Likewise, forgery is often employed in remixing news and other media content to promote an alternative narrative or interpretation of facts. Even where edits are not particularly hard to catch – such as the now-famous footage of White House reporter Jim Acosta that was slowed down at a single point to make it appear as though he pushed an intern (Martineau 2018) – there is often plausible deniability found in the general inability of the median reporter and news consumer to see minute changes (particularly in the absence of an original version).

DeepFakes resolve a key restraining factor in the production of digital forgery – the need to manually produce and test the provenance of forged media. By contrast, DeepFake image and video content is developed using a pair of neural network models. Initially, a face is studied by the algorithm and compared with the face of another person making analogous expressions and movements. The more training data that the algorithm has to

go off of, the better the results of this process are. To produce accurate fabrications of one face such that its attributes might be applied to the other, the algorithm generalises about each face in broad terms. Doing so allows for the generation of more believable synthetic facial content than might be the case if the programme simply tried to clone particular expressions onto a second face.

At this juncture, the second neural network goes to work. This algorithm is not generative but rather is aimed at identifying fabrications. The use of two algorithms in this way is a relatively new development called generative adversarial modelling (or, taken as a single model, a generative adversarial network (GAN)) (Goodfellow et al. 2014). Using the output of the fake-detecting network, the generative portion of the programme does not stop producing higher and higher fidelity fabrications until it is probabilistically confident of spoofing its counterpart (Shen et al. 2018). Thus, DeepFakes are not simply the products of advanced content production techniques – they are naturally adaptive and resilient to what we might think of as efforts to inoculate society by simply building better methods for their detection. Given that current methods for detecting DeepFakes rely largely on flaws in the way that algorithms execute their task – for instance, by examining discrepancies in blinking patterns (Li, Chang, and Lyu 2018) under the assumption that current programmes often don't have enough data to perfectly generalise about such nuanced behaviour – it seems an unsafe assumption that such techniques might work in the future. As one researcher put it, 'at some point it's likely that it's not going to be possible to detect [AI fakes] at all' (Vincent 2019).

At the time of writing, the internet is replete with examples of high fidelity DeepFake content. Remarkably few good quality DeepFake products have received widespread attention in news reporting or on major content aggregation sites like Reddit, but the number is growing rapidly. The digital media company *BuzzFeed* grabbed headlines when it teamed up with Jordan Peele to create a DeepFake video of former President Barack Obama in which Peele spoke about the threat of DeepFakes and illustrated – by making it appear as though Obama was calling President Trump a 'dipshit' – the potential of the technique (Silverman 2018). In entertainment, amateur producers have altered classic films, for instance convincingly placing the features of actors Robert Downey Jr. and Tom Holland on *Back to the Future* characters and even upgrading a CGI rendering of Carrie Fisher in the film *Rogue One: A Star Wars Story* to be more lifelike. More insidiously, video of political officials have been altered or fabricated to imply some problem or offence. Media of House of Representatives Speaker Nancy Pelosi was slowed down (The New York Times 2019) and used by trolls to imply an underlying medical issue, for instance, while video footage of President Donald Trump has consistently been altered to suggest a tirade against one or other interest group. While most such fabrications are flagged as such reasonably quickly (or are not intended to be taken seriously to begin with), these examples highlight a concerning reality – that while large volumes of input data (i.e. legitimate imagery and video content) is required to make convincing DeepFakes, that data is typically already widely available for public figures and is increasingly available for the median user of social internet platforms.

2.2. Are DeepFakes game-changing?

In the section below, I discuss the societal effects and public policy implications of deep learning AI techniques to the disinformation enterprise broadly. This discussion is valuable

in itself as a broad consideration of the manner in which disinformation capabilities, whether with individuals or sophisticated political actors, have been upgraded and continue to be upgraded in the age of the internet and advancing machine learning techniques. But is the specific form and method of DeepFake multimedia content itself a game-changing development that substantially alters the character of disinformation in societal perspective?

Below, I describe how the simultaneous democratisation of the technology and advancing fidelity of DeepFake products are the main points of concern going forward. Certainly, at present, there are substantial obstacles to the creation of good DeepFakes beyond media featuring prominent societal figures and by most amateur producers, namely that doing it well requires (1) immense and varied data on the target of a given fabrication and (2) a non-trivial amount of processing power. However, such obstacles are likely to fade away as all societal strata are increasingly recorded as a matter of course across common social, educational and consumer activities. Likewise, processing power continues to strengthen over superconductor generations and future services focused on enabling construction of DeepFake content will likely offer cloud support dedicated to the task. Moreover, as is the case with fake news more broadly, fabricated content need not be perfect in order to skew media representations and popular debate. Given this, I argue going forward that DeepFakes should be thought of as an evolution – rather than a revolution – in disinformation techniques, but one that nevertheless offers unique opportunities for disruption as the ability to rapidly create fake media of *reasonable* fidelity (i.e. defensible but not undetectable) at scale enables organised influence campaigns more dynamically capable of shaping discourse than has ever previously been the case.

3. The effects of AI-enabled disinformation

If, as much media reporting has suggested, fake content is poised to become virtually indistinguishable from the real thing, what can policymakers and other relevant stakeholders do to safeguard democratic, economic and other societal processes? To answer this question, it is first critical to imagine the implications of DeepFakes for the national security and political integrity of Western nations in greater detail than has been done to date. In this section, I address these implications and possible effects by referencing the coming prevalence of DeepFake content at several different levels – (1) at the level of the psyche of everyday consumers of media information, (2) as a highly tailored instrument that can be used to enhance the effect of other criminal or conflict actions, such as cyber operations, and (3) as a mass-produced, regular feature of the information environment in democracies. I also note the possibility of (4) compounding second and third-order effects that are less easy to foresee.

3.1. Human-level effects

Though it almost goes without saying, the clearest effect of a future in which DeepFakes are as commonplace in the lives of most Western citizens as is unaltered reporting is likely to be the continuing degradation of electorates' ability to remain informed and vigilant of such aberrations as might infringe upon social and political process. Recent scholarship

and investigative journalism have both shown the manner in which widespread exposure to fake news is having a rising impact on the cognition of the median voter in North America and Europe. Comparison of fake and factual news media reporting since 2016 by one publication showed a stark difference in the popularity of fabricated content over real (Silverman 2016). Yet another by researchers at MIT showed a similar pattern – fake news travels faster across internet platforms and spaces than does other content (Vosoughi, Roy, and Aral 2018). These trends may be expected if one assumes that fake content aims to ‘capture’ users with provocative or sensational titles. And yet subsequent research has initially shown that this trend holds true even where the focus of reporting in both categories remains relatively neutral (Pennycook, Cannon, and Rand 2018). In this way, it is easy to see how an intensification of the challenges surrounding validation of information and the attribution of data therein stands to introduce new public susceptibilities for misdirection and external influence.

One particular facet of human-level responses to fake news as a regular feature of an individual’s information environment bears notable consideration. In addition to a natural predilection towards simplicity in messaging that favours attempts to dissemble, some studies have demonstrated a tendency among humans towards the discounting of even apparently more credible reporting given a sufficiently saturated information environment or strong enough sociopolitical priors (ibid). Perhaps most worryingly, direct exposure to fake news muddies the ability of people to accurately identify the hallmarks of fabrication in subsequent viewings (ibid). This holds over and above partisan predispositions towards believing certain fake content to be credible given extremely low thresholds of quality assurance on the part of the party making it, explained by researchers as a lack of inclination to reason when viewing information that reinforces deeply seated beliefs (Clayton et al. 2019). Moreover, the effect intensifies among people with diminished cognitive abilities, who, alongside partisan or issue-invested populations, are among the most regularly targeted by prosecutors of influence campaigns (Pennycook and Rand 2019; Roets 2017).

With DeepFakes, the mere notion that content fabrication could become more tangible and visceral in the delivery is not necessarily enough to support a claim that disinformation will get anything more than incrementally more effective. Where DeepFakes *do* portend substantially enhanced opportunities for disruption is with the production of content that is either excusable or inflammatory purely on the merits of the media piece itself (rather than on the credibility of the messaging). Even where it is seen to be fabricated, fake content can be used to support conspiratorial narratives. And the fact of some imagined melding of personas, issues or symbols is enough itself to cause offence on a case-by-case basis. Indeed, high fidelity fake content is already the stuff of outrage in the limited instances where, for instance, celebrity faces have been placed on pornstar bodies to fabricate a sex scene or where politicians have been modelled such that they say shocking things. Artificial intelligence-aided fabrications that are increasingly cheap and accessible stand not only to worsen such outrage, but also potentially to induce fake news malaise and a coarsening of the moral underpinnings of most Western democracies. Regardless of eventual effects, the clear public policy questions become ones of rights of expression and of cumulative effects on the nature of sociopolitical discourse. I address both below.

3.2. Adjunct modification

Though DeepFake technology is arguably incremental in its upgrading of disinformation possibilities, the opportunity for a dramatic amplification of trends we've seen in recent years in the changing psyche of the average citizen is real given the potential pervasiveness of such rich and soon-to-be commonplace content. That said, I argue here that the greater threat to Western societies comes from the manner in which AI-enabled multimedia disinformation works at higher levels to (1) amplify the effects of other conflict processes and (2) broadly undermine the normative basis of democratic process. I address the first of these in this section.

When considering the prospective value of DeepFake and related AI-enabled forms of multimedia disinformation for organised crime and conflict actions, it is perhaps best to start by acknowledging that propaganda or coercive signalling rarely occur as standalone actions. Rather, such messaging most often plays a distinct role in *campaigns* designed to achieve some sort of strategic outcome, whether or not the objectives thereof are defined by terrorist enterprise, state interests or other machinations. For purposes of elaboration, I consider the particular shape of cyber campaigns wherein cyber operations are often utilised alongside disinformation, misinformation, counter-command-control intelligence actions and more to generate some form of strategic gain.

As Smeets argues, cyber operations need not entirely be understood as standalone incidents with static temporal effects and implications (Smeets 2018). Some cyber operations *enable* other cyber actions that allow an actor to further a strategic objective. This notion is well understood by cybersecurity experts who accept that seemingly unrelated cyber intrusions might be linked or conducted in aid of a broader cross-domain campaign. At the same time, some cyber operations *reinforce* other cyber actions. This is, to some degree, simply a non-linear form of enabling actions wherein multiple cyber actions combine to create an effect that achieves a strategic objective (as opposed to cyber actions that enable a subsequent, strategically meaningful one). And finally, cyber actions can *enable non-cyber* actions that achieve a strategic objective, such as when the digital theft of site logs leads to the criminal interdiction and indictment of a narcotics trading operation.

This conceptualisation of cyber attacks as complex campaigns is generalisable to other conflict processes – i.e. where any given kind of conflict action can (1) enable another similar action, (2) reinforce the effects of another similar action or (3) either enable or reinforce an entirely different kind of conflict action – and naturally applies to the directed employment of disinformation. Cyber operations might retrieve sensitive government, political or personal information that can then be leveraged to strategic effect in an influence campaign. Russian penetration of Democratic National Committee (DNC) systems in 2016 fuelled the strategic publication of embarrassing information about the Democratic Party's operating intentions and procedures. Importantly, partial information about the hack of the DNC lent credibility to malicious leaks and likely added weight to fabricated content being pushed across a range of mediums throughout the same period. This, as scholars, journalistic investigators and practitioners have noted in a diversity of reporting, likely created a loop wherein partial knowledge of cyber operations, some real information and much falsified content reinforced the credibility of the influence effort as a whole. In many of the most recent influence operations conducted by the

Russian Federation, not only were cyberattacks employed to gain private information for purposes of doxing to amplify the impact of fabricated content publication; rather, Russian-backed operators made extensive efforts to cultivate conspiratorial communities and varied their disinformation strategies so as to best build the narrative foundations that would ultimately serve to sustain broader rhetorical trends.

With DeepFakes, the risk is quite simply that such combining effects will be more easily obtained and sustained, both by prosecutors of influence campaigns targeting democratic systems and even by sociopolitical actors within Western civil society attempting to agenda set. This is not simply a matter of improved fidelity of fake news productions either, else we might simply come to the conclusion that DeepFake content yet again only incrementally upgrades the toolkit of influence. Such an argument is not necessarily obvious on the technical merits of the technology, even if the adversarial methods and resultant productions are impressive.

Rather, the risk lies with the fact that DeepFakes add no small amount of dynamism to the ability of sophisticated propagandists to shape. In the international relations (IR) literature, political actors are often envisioned as principally affecting their environment by either signalling or shaping. Signalling actions add context to strategic interactions via direct or indirect engagement with other actors. Shaping actions are those designed to *define* the environment in which such interactions occur, much as a corporation might privately lobby lawmakers to ban a chemical found in a competitor's product so as to improve the strength of marketing signals (i.e. advertisements). Disinformation succeeds, when employed to shape rather than to signal, when it either improves the utility of other conflict actions or diminishes the functionality of an opponent's capabilities. This implies a set of efforts more fragmented and diverse in their presentation than the notion of high fidelity fake content might suggest.

Functionally, while DeepFakes themselves may not worsen the security challenge for states, their use can either create or constitute second-order effects that aid broader interference efforts. DeepFake content released by anonymous sources following a public admission of data breach by a company or political party, for instance, is instantly more credible than multimedia released without similar priming of the public. A political party that experiences a server breach may find that DeepFake content of employees expressing racist sentiment about a candidate difficult to dismiss without a full reveal of the content of internal documentation. In the meantime, news cycles that might otherwise be consumed by legitimate social or political discourse are consumed. And DeepFakes utilised following the release of genuine compromised information adds value to an interference campaign, particularly if released alongside other less sophisticated fake content and even if the only resultant effect is a lengthening of the exposure period for the victim.

Normalising the use of DeepFakes in this way extends the notion that we are now in an era of few secrets in an interesting way. Not only is private information relatively insecure in comparison with in eras past; rather, the burden of proof is increasingly shifting to potential victims to demonstrate what private information they *aren't* protecting. Naturally, this shift contains numerous implications for foreign policy relations, where high quality DeepFake content of members of a country's leadership discussing possible military actions might dramatically change the tone and focus of international rhetoric during crisis episodes.

3.3. Environmental and compounding effects

Adding to the challenge of DeepFake content enabling other conflict processes, public policymakers must inevitably deal with the direct effects of high fidelity fabricated content as they manifest at scale within democratic states. In other words, the cognitive impact of pervasive DeepFake news is not simply something that manifests in the mind of the average citizen, but rather something that has distinct implications for the function of political systems that rely to some greater or lesser degree on the deliberations and interaction of those people. Simply, as a number of recent works have turned to argue, democratic political systems are perhaps best thought of as information systems (Farrell and Schneier 2018). In political science and in studies of political communication, a range of seminal theories contest different perspectives on how democracies work. Most focus on voting, noting either that citizens vote for those that will most closely represent their interests or that voting functions as a sort of adjudication mechanism for past events, with representatives more being judged on their track record than on their promises. Yet another more recent entrant to the literature explains that individuals vote as informed by social group identity. In each case, however, whatever format of preference voting plays out is ultimately informed by an underlying information ecosystem that moves popular deliberations on public and foreign policy issues towards moderate outcomes (e.g. Downs 1957; Snyder 1991; Russett 1994; Van Evera 2013; Reiter and Stam 2002) via the processing of information in an environment of institutions – from political representatives and media watchdogs to experts and interest groups – that infuses discourse with diverse perspectives.

Already, emerging scholarship suggests that sophisticated digital age information warfare campaigns are defined by an ability to interfere with this ecosystem of information assurance in substantial fashion. It is only natural to think that AI-enabled methods of multimedia fabrication will make such interference at scale systematically worse, as both the increasing ease of access to tools for creating DeepFakes and the rising fidelity of such products portend a threat to the fundamental information security of democratic polities. In fairness, it would be disingenuous to simply argue, out of step with the arguments made so far in this paper, that DeepFakes portend a blanket worsening of the challenge facing public policymakers. Rather, the most pronounced exacerbations of challenges faced by democracies in the digital age from AI-enabled multimedia disinformation likely manifest in specific changes to the capabilities of citizens – in this case, to discern and analyse information – rather than to the capabilities of propagandists. DeepFake technologies offer new opportunities, for instance, for bringing fringe perspectives before mainstream viewership as conspiratorial terminology or imagery can be more intimately inserted into shared status quo scenarios involving popular figures than ever before. Moreover, and of particular note-worthiness for policymakers, DeepFakes stand to produce second and third-order effects beyond the direct impacts discussed here that may dilute the character of democratic discourse and affect attitudes towards the credibility of such systems among citizenry.

4. DeepFake policy issues: technical and strategic solutions

Creating policy to deal with the negative externalities and more direct threats to democracy posed by AI-enabled multimedia disinformation is in some ways no different than is

the development of means for effectively combating fake news. There are added complexities involved, to be sure, but the threat faced by modern societies is simply a more intense version of the realisation episodes experienced by Western nations in the past several years focused around the issue of cyber-enabled subversion. This section explores legal, technological and normative approaches to countering insecurity emerging from the rise of fake content dissemination.

4.1. Regulating fake and DeepFake content

Before considering the technological and strategic dynamics of prospective efforts to address the challenge of DeepFakes, it is worth addressing the legal element. At the surface, the simplest solution to many issues of DeepFake content distribution and subsequent effect is efficient determination of what is and is not fake. However, as some others have initially pointed out in the context of more pedestrian fake news, regulation in such a vein is likely to face challenges on a number of fronts (Timmer 2016). First, the codified ability of any government entity to determine the ‘truth’ of any content leaves democratic processes open to partisan political manipulation. Unlike fact, truth is a relative concept and naturally varies across the interpretations of relevant operatives of the process even where the application of political bias is not overt. Where bias is overt, regulatory mechanisms of truth determination might easily be employed for political gain. A particularly worrying element of such a prospect is the co-option of protected elements of democratic process, particularly the judiciary, law enforcement and intelligence.

Second, for any effort to regulate fake news at any level of sophistication there exists the inevitable challenge of ensuring robust and consistent results in the determination of fact. Particularly where artificial intelligence comes into play to enhance the capacity of sociopolitical actors to fool even attentive parties, the likelihood that a substantial fraction of fact determination would result in false positives/negatives chances a slide towards censorship. After all, faced with a non-trivial failure in any regulatory ability to expunge fabricated content from democratic information environments, regulators may be inclined to encourage media gatekeepers to err on the side of restrictiveness.

And third, perhaps most importantly, regulatory efforts in this vein are virtually destined to clash with protections placed around the speech of citizens, which are commonly already set up to protect false speech in the West. In the United States, in particular, the unconstitutionality of anti-disinformation legislation aimed at the rising rides of fake news is nigh determined by precedent (Hundley 2017). As Timmer (2016, 678) points out, the Supreme Court often holds up political speech above other forms of speech as most deserving of First Amendment protections. Not only is this intention to ensure political freedom an obvious foil to efforts to regulate speech in any form; rather, the Supreme Court has explicitly stated a protection for false speech under the First Amendment. In *Cantwell vs. Connecticut*, the Court held up the idea that the First Amendment has no value propositions enshrined or implied therein. ‘Truth’ is not an applicable concept under the law in this case. As the Court stated in *Cantwell v. Connecticut* (1940):

In the realm of religious faith, and in that of political belief, sharp differences arise. In both fields the tenets of one man may seem the rankest error to his neighbour. To persuade others to his own point of view, the pleader, as we know, at times, resorts to exaggeration, to vilification of men who have been, or are, prominent in church or state, and even to

false statement. But the people of this nation have ordained in the light of history, that, in spite of the probability of excesses and abuses, these liberties are, in the long view, essential to enlightened opinion and right conduct on the part of the citizens of a democracy. (310 U.S. 296)

This opinion creates firm precedent for the notion that false speech is protected by the First Amendment, including in cases of error, sociopolitical, religious or other interpretation, or intentional political spin.

A legal option for combating the influence of DeepFakes that does not rely on direct regulation of speech, of course, is essentially to rely on the fact that speech occurring on private platforms supported by technology companies does not receive full protection under constitutional. Speech that takes place on Facebook, Twitter or similar sites is not public speech and is, therefore, more clearly open to regulation on the part of the company involved, which has occurred somewhat in recent years as companies respond to public outrage.

The most obvious recent incidence of this has been the steps taken by Facebook, Google, Twitter and others in the wake of Russian activities revealed to have taken place at some scale on their platforms. Facebook and Google both announced major changes to reduce the visibility of sensational or spam content (Hoffmann, Taylor, and Bradshaw 2019). Google has, in recent years, heightened the required weight of factors required to push results to the top of web search queries, including keyword density, site age, site certification, site traffic and PageRank metrics related to links embedded in a given webpage. Facebook, on the other hand, has more directly attempted to tailor social interactions towards 'meaningful' engagement, promoting specific formats of inter-user engagement and massively de-weighting the value of content that encourages active promotion of voting, share or comment features (e.g. "Like" and "share" this post if you love the Washington Nationals'). As one report notes, however, these measures have often been far less limited in scale of implementation than suggested and bring yet additional opportunities for manipulation. Hoffmann, Taylor, and Bradshaw (2019) note that Facebook's actions, for instance, keep users on the platform more consistently, which has the effect of allowing clear methods for users to improve their algorithmic reputation and to be able to push content far and wide regardless of its quality.

With regard to the legal dimensions of mitigating AI-enabled disinformation, government approaches to combating sophisticated fake news might target the sources thereof. One advantage of the oncoming sophistication of fake news creation is the codification of production capabilities in software. While many have expressed some measure of fear that the proliferation of software for creating DeepFake content is a democratisation of malicious potential mirroring the global spread of hacker kits over the past several decades, it is also the case that more formal tools for fake news production give law enforcement and judicial forces a foothold from which to base prohibitions.

An early example of how formal manifestation of such toolkits might play into government hands exists in the form of existing fake news content marketplaces (Gu, Kropotov, and Yarochkin 2017). Already, the underground market for sale of software designed around malicious intent is rife with service and product offers centred on the fabrication of both content and influence. Service providers like the Chinese site Xiezuobang (写作帮) or sellers found on Weixinvips offer basic production of fabricated text content or simplistic vote manipulation, while others like the Public Opinion Influencing System (博约舆情

引导考评系统) and the Yunjing Public Opinion Monitoring System (天互云镜舆情监测分析系统) offer more holistic services focused on rapidly boosting the influence and exposure potential of client content creators. Many Russian sites have sprung up in just the past decade to mimic the hacker marketplace emphasis on pre-packaged software – such as social media spamming scripts for the unsophisticated user – and have themselves been mimicked in the form of sales forums in South Asia and the Middle East that specialise in tools that do everything from automate account creation to streamline fake news production. Such marketplaces are not uncommon in English-speaking parts of the web either and present as a distinct opportunity for interdiction of sophisticated fake news production.

Ultimately, however, it is important to this paper's analysis of AI-enabled disinformation specifically to note that each of these challenges and avenues of opportunity are substantially *the same* as those that pertain to contemporary manifestations of fake news. This suggests that litigative approaches to combating malicious use of DeepFakes is likely to be successful only where clear cases of libel emerge or in the takedown of criminal markets for restricted products. Otherwise, efforts to control such disinformation as speech is far more likely to come down to the operation of private online spaces like Facebook and Twitter than it is to core challenges on the constitutionality of such fabrications.

4.2. Technology's role in neutralising insecurity from disinformation

Most initial proposals on the combating of AI-enabled multimedia disinformation employed for malicious purpose tend to turn to one of two technological solutions – (1) detection of fabricated or modified content and (2) 'audience empowerment' solutions. On the first solution, with AI-enabled disinformation approaches to detection run into the problem of adversarial machine learning as the basis for much multimedia fabrication. The entire point of GANs that produce fake content is to 'beat' a detection algorithm. Though there is the not-insignificant matter of implementation to consider, any significant advance in detection notionally promises to be a corollary of better DeepFake production techniques. This issue is only likely to become more apparent as programmes outpace current shortcomings borne of insufficient training data. After all, most current detection techniques work around the principle of finding the outputs of flawed programming assumptions or imperfect implementation (e.g. the discovery of repetitive motions, imperfect mimicry of human blinking, etc.). As such issues become less commonplace in fabricated content intended for malicious effect, both the margin of error in asserting the 'fact' of multimedia and the lifespan of abilities to so assert such 'fact' shrink dramatically.

With regard to 'audience empowerment' (Levi 2017), a promising solution for widespread dissemination of high fidelity fake content would be any effort that encourages or requires media outlets and social platforms to police their content with some systematic rigour. As noted above, major technology companies in the United States might be successfully coaxed away from insubstantial action emerging from CDA 280 protections if they are convinced that the financial and reputational costs are significant. However, the focus of governments and public interest groups concerned about sophisticated fake news content need not be blanket censorship thereof. Constitutionally, government stands on a significantly stronger leg when it comes to legislation that prescribes the formats and extent of information delivery required by media stakeholders so long as

the focus is annotative and not restrictive in nature. Naturally, empowering audiences to be critical of the nature – not just the substance – of the content they are presented is a desirable feature of the information environment even in a world where fake news is not pervasive. But an annotative solution to fake news has the advantage of not organically invoking either public or private actors in the role of the censor. And, added to the attraction of this approach, it is something clearly desired by Western populations (Horrikan and Gramlich 2017).

The main issue with most suggestions of audience empowerment as a foil to sophisticated and widespread disinformation in society lies in the implied role of gatekeepers that would have to offer discrete judgment on the providence of content. In short, few want Facebook to be the arbiter of content quality even if they object to annotation as a government service. There will undoubtedly be decentralised options for services that can be plugged into platforms across the web to achieve such annotation aimed at audience empowerment. Relying on crowds to either generate quality ratings along several lines or train algorithms to produce the same might democratise the process of protecting democratic information processes and thus award such a solution credibility. The categories described above in the description of democracies as information systems – i.e. the origination of data, the credibility thereof, etc. – are a good starting point for imaging the kinds of metadata needed to assure information security and protections from disinformation at scale.

4.3. Pathways to deterrence and constraining norms

A final policy challenge centred on the continued sophistication of disinformation is the attractiveness of its employment for the purposes of subversion and disruption. In particular, many countries have incorporated fake news into their adversarial operations targeting Western power and influence. In the age of artificial intelligence, it seems likely that this use of disinformation to enhance hybrid warfare on a number of fronts will continue.

Subversion, and more specifically the employment of disinformation, is difficult to deter in any meaningful way because it is not fundamentally a strategy of contestation. Rather, subversion is a strategy of disempowerment.⁹ Whether the intention is interference designed to induce cognitive dissonance among societal adversaries or some broader effort to erode the foundations of interests an actor deems threatening to their own, deterrent signalling about potential costs of continued activity is almost destined to suffer from two things. First, because the aim of much organised use of disinformation is erosion of power, foreign action is most likely to be found where there is limited opportunities for triggering more well-established thresholds of escalation. This implies an enthusiastic willingness to disengage when discovered and makes cost imposition a hard sell. Second, both the attack surface and attack vectors involved in sophisticated information warfare campaigns are extensive. Because of this, signalling urgently, credibly and clearly about what might happen if foreign adversaries continue their employment of disinformation is extremely difficult. Much as is now assumed about deterrence of cyber conflict actions, counter-offensive action or threats thereof will need to occur rapidly and must communicate intention in reasonable detail, something that is fundamentally difficult to do given the mediums of engagement involved.

So how might the United States and other Western nations effectively dissuade countries like China, Russia, Iran, Saudi Arabia and North Korea from continuing to employ disinformation in more and more sophisticated, widespread fashion? Many of the defensive options described above, particularly the construction of an audience empowerment ecosystem, may help in some small measure. And yet, again as with cyber conflict, robust defence against societal disinformation would need to be exquisite in order to produce a broad deterrent effect. Likewise, while massive criminalisation of the use of certain tools for creating DeepFakes among American and European partner nations might produce norms of non-use in the Western countries, such mechanisms for norm generation are unlikely to significantly affect non-Western societies. Rather, it is much more likely that deterrent success will emerge from a blended strategy of entanglement and informational brinksmanship.

Entanglement is any deterrent strategy that ties the costs involved in an attack to the welfare of the prospective attacker. In broad terms, for instance, economic warfare between China and the United States is generally always likely to produce mutual costs and is thus traditionally a less attractive option for either side. With disinformation, constraining norms of non-interference centred on the sophisticated use of fabricated information might emerge from counter-offensive information warfare and efforts to empower populations under authoritarian regimes with access to such tools, particularly with countries like Iran and China that value non-interference in domestic affairs. A persistent enough tempo of mutual engagement may, as has been suggested with operations in cyberspace more specifically, aid efforts to communicate punitive intention and produce deterrent learning based on tacit understandings of the role of disinformation in securing state interests.

5. Conclusion

In large part, the challenge of AI-enabled multimedia disinformation is evolutionary, not revolutionary. The advent of easy-to-produce DeepFakes and related content is simply a continuation of the fake news challenges that are not uncommon in world history and have particularly characterised the first years of the twenty-first century. What artificial intelligence *does* do is simultaneously democratise and sophisticate the production of high fidelity multimedia fabrication. In so many ways, AI-enabled multimedia as fake news promises to make systematic effects that have up until now been felt by Western societies in a punctuated fashion.

This essay argues, above all else, that the emerging debate on DeepFakes and on artificial intelligence on this front does not place enough emphasis on these systematic effects. While there is certainly danger to Western societies bound up in the individual- and group-level effects of DeepFake content, these challenges scale to have serious implications for the function of the median democratic polity. With fake news, scholarship has already demonstrated a tendency towards perspective hardening, malaise and the development of deep-seated mistrust of even credible information where an information environment is sufficiently saturated with mixed content. Such tendencies are deleterious to the consensus-garnering, fact-finding function of democratic information systems. Moreover, they stand to compound the opportunities for new instability that comes from blending high fidelity content fabrication with other modes of hybrid warfare engagement.

The most effective ways to deal with the advent of AI-based fake news as a continuation of today's crisis of disinformation are not regulatory so much as they are about building

democratic resilience and generating constraining norms around information warfare. Organised effectively, public pressure can be brought to bear against technology and media companies as the most significant stakeholders of the modern global information environment. At the same time, Western governments must more proactively internalise the notion that foreign interference is an unavoidable reality of interstate conflict in the twenty-first century. Moreover, since subversion is not fundamentally about contestation and as much use of disinformation is about subversion, Western governments would also do well to consider the offensive use of information operations and active support of foreign dissidents as a mechanism for imposing costs on prospective belligerents in this vein.

That said, the debate over artificial intelligence and disinformation is just beginning. This essay has offered context for policymakers interested in thinking more completely about the scope of the challenge posed by DeepFakes and related content. While I have suggested pathways for potential threat mitigation, policy in this area will inevitably need to thoughtfully balance the competing imperatives of citizens' rights, the functionality of democratic processes and the national security challenges bound up in advancing means of waging conflict beyond the realm of traditional hostilities.

Notes

1. Though the practice of fabricating news content is age old, the term 'fake news' re-entered the popular lexicon in force in 2016, originating with the efforts of troll farms in the Macedonian town of Veles. See Wendling (2018).
2. Specifically, this essay regularly employs the term 'AI-enabled disinformation' to refer to the prospective ubiquity of DeepFake usage beyond the already-commonplace basic machine learning utilised in disinformation campaigns today.
3. On IW more broadly, see *inter alia* Badawy, Ferrara, and Lerman (2018); Zannettou et al. (2019); Stewart, Arif, and Starbird (2018); Stukal et al. (2017).
4. For perhaps the most cohesive description of trolling activities in one such campaign, see Jamieson (2018).
5. Marchal et al. (2019) label this immense body of material 'junk news'.
6. The experience with well-organised, Web 2.0-enabled disinformation from at least 2014 onwards has led to a resurgence of debate about terms like 'post-truth' or a 'post-factual' world. On this, see *inter alia* MacMullen (2020); Hendricks and Vestergaard (2019); Durnová (2019); Hendricks and Vestergaard (2018).
7. Admittedly, this argument is technologically deterministic and subject to reasonable push-back. Benkler, Faris, and Roberts (2018), for instance, argue that the crisis of truth and democratic process that is apparent in the United States is based on a series of architectural and cultural factors present in the underlying fabric of the American nation. And yet, even in such critiques of the broad assertion that new media systems and technologies are responsible for the incipient crisis there is an argument made that the technology is a significant – indeed, the most significant – intervening factor responsible for amplifying pre-existing inclinations. Therefore, while the argument that technology alone does not make a crisis is a compelling one, it is also hard to ignore as holding at least some water. Also see Page and Gilens (2020); Grinberg et al. (2019).
8. For one of the best broad descriptions of computer development in the context of modern advances in machine learning and reasoning, see Domingos (2015).
9. Subversion, though generally understudied by scholars in the social sciences, has received some notable treatment in recent years in the wake of digital age influence campaigns. For reference, see *inter alia* Beilenson (1972); Kahin and Kahin (1997); Whyte (2017); Breitenbach and Byrjalsen (2019); Zilincik, Myklin, and Kovanda (2019).

Notes on contributor

Dr. Christopher Whyte is assistant professor at Virginia Commonwealth University in the L. Douglas Wilder School of Government and Public Affairs' program on Homeland Security and Emergency Preparedness. He teaches coursework on cybersecurity policy, cyber conflict dynamics, political risk analysis and strategic planning. His research interests include a range of international security topics related to the use of information technology in war and peace, political communication and cybersecurity doctrine/policy. His published work, including two books and numerous articles, examines the contours of digital age information warfare, decision-making in cyber operations, online fringe social spaces and artificial intelligence.

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

- Abrams, Steve. 2016. "Beyond Propaganda: Soviet Active Measures in Putin's Russia." *Connections* 15 (1): 5–31.
- Badawy, Adam, Emilio Ferrara, and Kristina Lerman. 2018. "Analyzing the Digital Traces of Political Manipulation: The 2016 Russian Interference Twitter Campaign." In *2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 258–265. IEEE.
- Beale, Mark H., Howard B. Demuth, and Martin T. Hagan. 1996. *Neural Network Design*. Boston: PWS Publishing Co.
- Beilenson, Laurence W. 1972. *Power Through Subversion*. Washington, DC: Public Affairs Press.
- Benkler, Yochai, Robert Faris, and Hal Roberts. 2018. *Network Propaganda: Manipulation, Disinformation, and Radicalization in American Politics*. Oxford: Oxford University Press.
- Boyd, Danah. 2017. "Google and Facebook Can't Just Make Fake News Disappear." *Wired*. Link: www.wired.com/2017/03/google-and-facebook-cant-just-make-fake-news-disappear.
- Breitenbauch, Henrik, and Niels Byrjalsen. 2019. "Subversion, Statecraft and Liberal Democracy." *Survival* 61 (4): 31–41.
- Cantwell v. Connecticut. 1940. 310 U.S. 296. US Supreme Court.
- Clayton, Katherine, Jase Davis, Kristen Hinckley, and Yusaku Horiuchi. 2019. "Partisan Motivated Reasoning and Misinformation in the Media: Is News From Ideologically Uncongenial Sources More Suspicious?" *Japanese Journal of Political Science* 20 (3): 129–142.
- DiResta, Renee, Kris Shaffer, Becky Ruppel, David Sullivan, Robert Matney, Ryan Fox, Jonathan Albright, and Ben Johnson. 2019. *The Tactics & Tropes of the Internet Research Agency*.
- Domingos, Pedro. 2015. *The Master Algorithm: How the Quest for the Ultimate Learning Machine Will Remake Our World*. New York, NY: Basic Books.
- Downs, Anthony. 1957. *An Economic Theory of Democracy*. New York, NY: Harper, 260–276.
- Durnová, Anna. 2019. *Understanding Emotions in Post-Factual Politics: Negotiating Truth*. Cheltenham: Edward Elgar Publishing.
- Farrell, Henry, and Bruce Schneier. 2018. *Common-Knowledge Attacks on Democracy*. Berkman Klein Center Research Publication 2018–7.
- Goodfellow, Ian, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. "Generative Adversarial Nets." In *Advances in Neural Information Processing Systems*, 2672–2680. <http://papers.nips.cc/paper/5423-generative-adversarial-nets.pdf>.
- Gregory, Sam. n.d. "Deepfakes and Synthetic Media: What Should We Fear? What Can we Do?" *Witness*. Accessed 14 April 2020. <https://blog.witness.org/2018/07/deepfakes/>.
- Grinberg, Nir, Kenneth Joseph, Lisa Friedland, Briony Swire-Thompson, and David Lazer. 2019. "Fake News on Twitter During the 2016 US Presidential Election." *Science* 363 (6425): 374–378.

- Gu, Lion, Vladimir Kropotov, and Fyodor Yarochkin. 2017. "The Fake News Machine: How Propagandists Abuse the Internet and Manipulate the Public." *Trend Micro* 5.
- Hendricks, Vincent F., and Mads Vestergaard. 2018. *Reality Lost: Markets of Attention, Misinformation and Manipulation*. Cham: Springer International Publishing.
- Hendricks, Vincent F., and Mads Vestergaard. 2019. "The Post-Factual Democracy." In *Reality Lost*, edited by V. F. Hendricks, and M. Vestergaard, 103–117. Cham: Springer International Publishing.
- Hoffmann, Stacie, Emily Taylor, and Samantha Bradshaw. 2019. *The Market of Disinformation*. Oxford: University of Oxford.
- Horrigan, John B., and John Gramlich. 2017. "Many Americans, Especially Blacks and Hispanics, are Hungry for Help as They Sort Through Information." *FactTank: News in the Numbers*. Washington, DC: PEW Research Center.
- Hotchkiss, Michael Bennett. 2017. "Nostradamus Prophecy as a Russian Information Warfare Concept." In *European Conference on Cyber Warfare and Security*, 172–175. Academic Conferences International Limited.
- Hundley, Annie C. 2017. "Fake News and the First Amendment: How False Political Speech Kills the Marketplace of Ideas." *Tulane Law Review* 92 (2): 497.
- Jack, Caroline. 2017. "Lexicon of Lies: Terms for Problematic Information." *Data & Society* 3. https://datasociety.net/pubs/oh/DataAndSociety_LexiconofLies.pdf.
- Jamieson, Kathleen Hall. 2018. *Cyberwar: How Russian Hackers and Trolls Helped Elect a President: What we Don't, Can't, and Do Know*. Oxford: Oxford University Press.
- Johnson, Boris. 2019. "Deepfakes are solvable—but don't forget that 'shallowfakes' are already pervasive." *MIT Technology Review*, March 25.
- Kahin, Audrey, and George McTurnan Kahin. 1997. *Subversion as Foreign Policy: The Secret Eisenhower and Dulles Debacle in Indonesia*. Washington, DC: University of Washington Press.
- Kaplan, Richard L. 2008. "Yellow Journalism." *The International Encyclopedia of Communication*.
- Kedar, Hadas Emma. 2020. "Fake News in Media Art: Fake News as a Media Art Practice vs. Fake News in Politics." *Postdigital Science and Education* 2 (1): 132–146.
- Levi, Lili. 2017. "Real 'Fake News' and Fake 'Fake News'." *First Amendment Law Review* 16: 232.
- Li, Yuezun, Ming-Ching Chang, and Siwei Lyu. 2018. "In Ictu Oculi: Exposing AI Generated Fake Face Videos by Detecting Eye Blinking." In *2018 IEEE International Workshop on Information Forensics and Security (WIFS)*, 1–7. IEEE.
- MacMullen, Ian. 2020. "Survey Article: What Is 'Post-Factual' Politics?" *Journal of Political Philosophy* 28 (1): 97–116.
- Marchal, Nahema, Bebece Kollanyi, Lisa-Maria Neudert, and Philip N. Howard. 2019. "Junk News During the EU Parliamentary Elections: Lessons From a Seven-language Study of Twitter and Facebook." *Oxford Internet Institute*.
- Martineau, Paris. 2018. "How an InfoWars Video Became a White House Tweet." *Wired*, November 8. <https://www.wired.com/story/infowars-video-white-house-cnn-jim-acosta-tweet/>.
- Page, Benjamin I., and Martin Gilens. 2020. *Democracy in America?: What has Gone Wrong and What we Can do About it*. Chicago, IL: University of Chicago Press.
- Pennycook, Gordon, Tyrone D. Cannon, and David G. Rand. 2018. "Prior Exposure Increases Perceived Accuracy of Fake News." *Journal of Experimental Psychology: General* 147 (12): 1865–1880.
- Pennycook, Gordon, and David G. Rand. 2019. "Lazy, not Biased: Susceptibility to Partisan Fake News is Better Explained by Lack of Reasoning Than by Motivated Reasoning." *Cognition* 188: 39–50.
- Reiter, Dan, and Allan C. Stam. 2002. *Democracies at War*. Princeton, NJ: Princeton University Press.
- Roets, Arne. 2017. "'Fake News': Incorrect, but Hard to Correct. The Role of Cognitive Ability on the Impact of False Information on Social Impressions." *Intelligence* 65: 107–110.
- Russett, Bruce. 1994. *Grasping the Democratic Peace: Principles for a Post-Cold War World*. Princeton, NJ: Princeton University Press.
- Shane, Scott, and Mark Mazzetti. 2018. "The Plot to Subvert an Election. Unraveling the Russia Story So Far." *New York Times*, September 20.
- Shen, Tianxiang, Ruixian Liu, Ju Bai, and Zheng Li. 2018. "'Deep Fakes' Using Generative Adversarial Networks (GAN)". San Diego, CA: University of California.

- Silverman, Craig. 2016. "This Analysis Shows How Viral Fake Election News Stories Outperformed Real News on Facebook." *BuzzFeed news* 16.
- Silverman, Craig. 2018. "How to Spot a Deepfake Like the Barack Obama–Jordan Peele Video." *BuzzFeed*. <https://www.buzzfeed.com/craigsilverman/obama-jordan-peelee-deepfake-video-debunk-buzzfeed>.
- Smeets, Max. 2018. "The Strategic Promise of Offensive Cyber Operations." *Strategic Studies Quarterly* 12 (3): 90–113.
- Snyder, Jack. 1991. *Myths of Empire: Domestic Politics and International Ambition*, 32. Ithaca, NY: Cornell University Press.
- Spencer, David Ralph, and Judith Spencer. 2007. *The Yellow Journalism: The Press and America's Emergence as a World Power*. Evanston, IL: Northwestern University Press.
- Stewart, Leo G., Ahmer Arif, and Kate Starbird. 2018. "Examining Trolls and Polarization with a Retweet Network." In Proc. ACM WSDM, Workshop on Misinformation and Misbehavior Mining on the Web.
- Stukal, Denis, Sergey Sanovich, Richard Bonneau, and Joshua A. Tucker. 2017. "Detecting Bots on Russian Political Twitter." *Big Data* 5 (4): 310–324.
- Sullivan, Margaret. 2017. "It's Time to Retire the Tainted Term 'fake news'." *Washington Post* 8. https://www.washingtonpost.com/lifestyle/style/its-time-to-retire-the-tainted-term-fake-news/2017/01/06/a5a7516c-d375-11e6-945a-76f69a399dd5_story.html.
- Sunstein, Cass R., and Adrian Vermeule. 2009. "Conspiracy Theories: Causes and Cures." *Journal of Political Philosophy* 17 (2): 202–227.
- Tandoc, Jr., E. C., Z. W. Lim, and R. Ling. 2017. "Defining 'Fake News': A Typology of Scholarly Definitions." *Digital Journalism*, 5 (7): 1–17.
- Times, New York. 2019. "Edited Pelosi Video vs. the Original: A Side-by-Side Comparison." *New York Times*, May 24. <https://www.nytimes.com/video/us/poli-tics/100000006525055/pelosi-video-doctored.html>.
- Timmer, Joel. 2016. "Fighting Falsity: Fake News, Facebook, and the First Amendment." *Cardozo Arts & Ent. L.J.* 35: 669.
- Van Evera, Stephen. 2013. *Causes of War: Power and the Roots of Conflict*. Ithaca, NY: Cornell University Press.
- Vincent, James. 2019. "Deepfake Detection Algorithms Will Never Be Enough." *The Verge*.
- Vosoughi, Soroush, Deb Roy, and Sinan Aral. 2018. "The Spread of True and False News Online." *Science* 359 (6380): 1146–1151.
- Wardle, Claire. 2017. "Fake News. It's Complicated." *First Draft News* 16. <https://firstdraftnews.com/fake-news-complicated/>.
- Wardle, Claire, and Hossein Derakhshan. 2017. "Information Disorder: Toward an Interdisciplinary Framework for Research and Policy Making." Council of Europe Report 27.
- Way, Lucan Ahmad, and Adam E. Casey. 2019. "How Can We Know if Russia is a Threat to Western Democracy? Understanding the Impact of Russia's Second Wave of Election Interference".
- Wending, Mike. 2018. "The (Almost) Complete History of 'Fake News'." *BBC News* 22.
- Whyte, Christopher. 2017. "Out of the Shadows: Subversion and Counterculture in the Digital Age." (PhD dissertation).
- Woolley, Samuel C., and Philip N. Howard. 2018. *Computational Propaganda: Political Parties, Politicians, and Political Manipulation on Social Media*. Oxford: Oxford University Press.
- Zannettou, Savvas, Tristan Caulfield, Emiliano De Cristofaro, Michael Sirivianos, Gianluca Stringhini, and Jeremy Blackburn. 2019. "Disinformation Warfare: Understanding State-sponsored Trolls on Twitter and Their Influence on the Web." In Companion Proceedings of The 2019 World Wide Web Conference, 218–226.
- Zilincik, Samuel, Michael Myklin, and Petr Kovanda. 2019. "Cyber Power and Control: A Perspective From Strategic Theory." *Journal of Cyber Policy* 4 (2): 290–301.
- Zuckerman, Ethan. 2017. "Stop Saying Fake News, It's not Helping." *My Heart is in Accra*. <http://www.ethanzuckerman.com/blog/2017/01/30/stop-saying-fake-news-its-not-helping/>.